



Stepper Motor for foam laser cutter

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Abstract

This study presents the implementation of a precise motion control system for a foam laser cutter using a stepper motor driven by an Arduino Mega and a TB6600 microstep driver. The objective was to enable controlled linear motion over a 30 cm rail using open-loop step signals. A NEMA-series 6-lead bipolar stepper motor was connected using Bipolar Series wiring configuration, and driven through pulse and direction signals generated via Arduino pins. A regulated 12V DC power supply provided the required voltage to the motor driver.

Experimental trials were conducted across a range of delay values (100 μ s to 6000 μ s) between step pulses, and the resulting displacement times were recorded. From this, velocity was calculated and plotted against delay. Results demonstrated a strong inverse non-linear relationship between the delay and the stepper motor's linear velocity. A second-order polynomial trendline was fitted to the experimental data, yielding a regression model with an R^2 value of 0.656, enabling analytical prediction of motion behavior for various delay inputs.

This system forms the foundation for an expandable motion platform. Future work involves scaling the system to control four stepper motors—two aligned horizontally and two vertically—to allow parallel motion in both axes. This will require evaluating multi-driver configurations and whether a single Arduino Mega can manage all signals simultaneously, or if distributed control will be necessary. The project demonstrates a practical approach to low-cost CNC positioning and offers a basis for extending into full 2D motion control in foam cutting applications.

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1. Introduction

Precise motion control is essential in modern manufacturing processes, especially in applications requiring high accuracy and repeatability. One such application is foam laser cutting, where even slight deviations in movement can affect cut quality and efficiency.

This project focuses on implementing a stepper motor system controlled by an Arduino to drive a foam laser cutter with consistent linear motion.

The objective is to:

- Integrate a stepper motor with an Arduino board
- Develop and upload control logic using Arduino IDE
- Achieve smooth, repeatable motion for foam cutting operations

This system allows cost-effective automation of movement along a predefined axis.

Unlike DC motors, stepper motors offer precise control over position and speed without feedback sensors. When paired with an appropriate driver and microcontroller, they provide an efficient, programmable solution for motion-based tasks like laser cutting.

2. Components Used

2.1. Power Supply

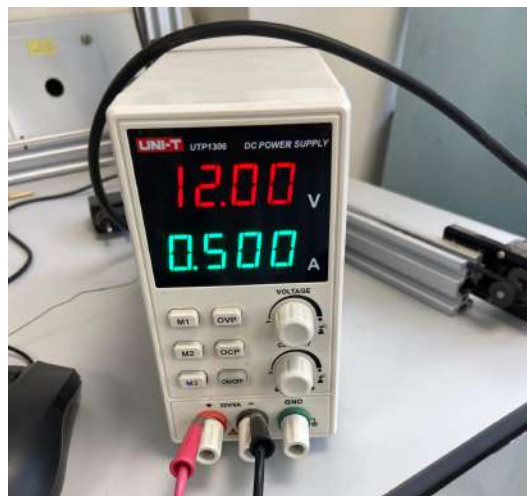


Fig1:power supply

2.2. Stepper Motor



Fig2:Stepper Motor

2.3. Microstep Driver (TB6600)

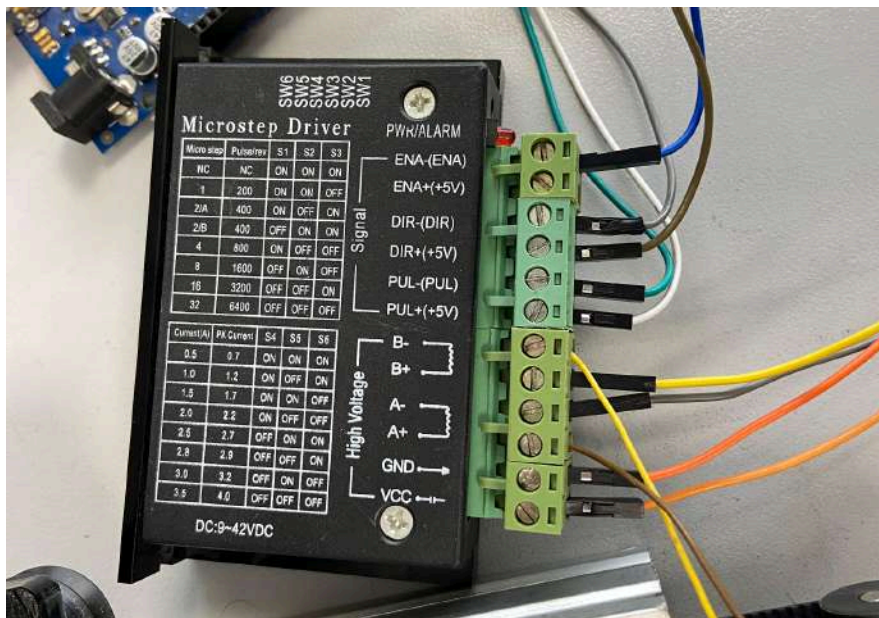


Fig3:Driver

2.4. Arduino Mega

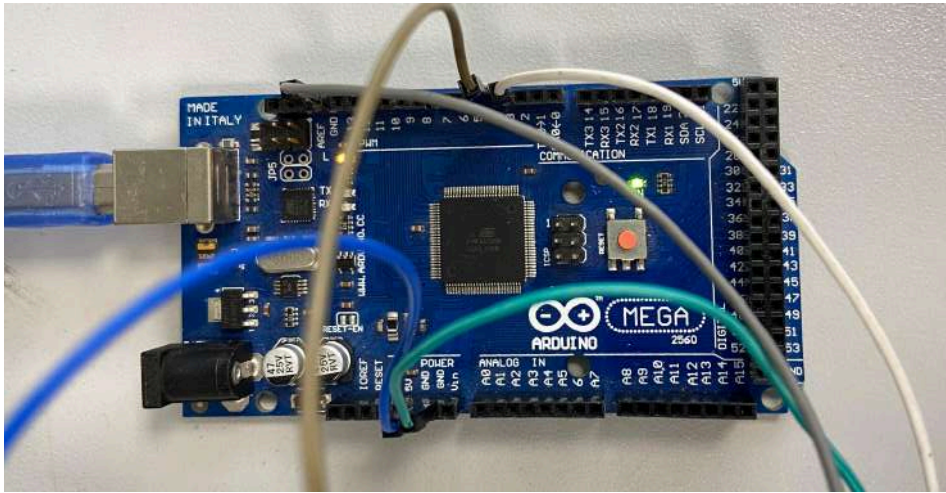


Fig4:Arduino Mega

3. Methodology

3.1. Wiring Diagram

A bipolar stepper motor with six leads was used. According to the datasheet, the appropriate configuration is the Bipolar Drive Series Connection, using Color Code 2:

- Brown, Black, Orange, Red, White, Yellow
- A coil: Brown (A+), Orange (A-)
- B coil: Red (B+), Yellow (B-)

The following diagram shows the wiring logic:

6 LEAD WIRES	1	2	3	4	5	6
Color Code 1	Red	White	Blue	Green	Yellow	Black
Color Code 2	Brown	Black	Orange	Red	White	Yellow
Color Code 3	Red	Black	Red White Stripe	Green	White	Green White Stripe
Bipolar Drive Half Coil Connection	A	$\bar{A}$		B	$\bar{B}$	
		$\bar{A}$	A		$\bar{B}$	B
Bipolar Drive Series Connection	A		$\bar{A}$	B		$\bar{B}$
Unipolar Drive	A	A/C Comm	C	B	B/D Comm	D

Figure 5: Wiring Color Code for Bipolar Stepper Motors

3.2. Our connection

The table below details the connections between the TB6600 driver, Arduino Mega, stepper motor, and power supply.

Signal	ENA-(ENA)	Ground (Blue Wire)	Driver to Arduino
	ENA+(+5v)	No connection	
	DIR-(DIR)	Ground (Grey Wire)	
	DIR+(+5V)	PIN 4(Brown Wire)	
	PUL-(PUL)	Ground (Green Wire)	
	PUL+(+5V)	PIN 3 (White wire)	
High Voltage	B-	Yellow Wire	Stepper to Driver
	B+	Red Wire	
	A-	Orange Wire	

	A+	Brown Wire	
	GND	Power Supply (Black Wire)	
	VCC	Power Supply (Red Wire)	

Table 1: Connection Table (Driver–Arduino and Stepper–Driver–Power Supply)

Steps to Run a Stepper Motor with Arduino

Steps to Run a Stepper Motor with Arduino for Foam Laser Cutter

1. Connect the Power Supply (Figure 1)
 - Set the DC power supply to 12V and limit the current to 0.5A
 - Connect the black wire to GND and the red wire to VCC terminals of the TB6600 driver
2. Wire the Stepper Motor to the Driver (Figure 2, Figure 3, Figure 6 ,7)
 - Identify the coil wires using color as in Figure 5
 - Connect:
 - A+ → Brown
 - A- → Orange
 - B+ → Red
 - B- → Yellow
 - Match these to the driver terminals as shown in Figure 7
3. Connect the Arduino Mega to the TB6600 Driver (Figure 4, Figure 6)
 - DIR+ → PIN 4 (Brown wire)
 - PUL+ → PIN 3 (White wire)
 - DIR-, PUL-, and ENA- → Arduino GND

- Leave ENA+ unconnected
- 4. Double-Check All Wiring (Figure 5, Figure 9)
 - Verify wiring color and connections using the wiring table and driver labeling
 - Make sure all ground connections are common
- 5. Upload the Arduino Code (attached in appendix)
 - Write code to generate step pulses on PIN 3 and control direction from PIN 4
 - Upload the sketch using the Arduino IDE
 - Stit switch at off on off on on on in the same order from sw1 to sw 6
- 6. Power the System and Observe Motor Response
 - Power on the 12V supply
 - Connect Arduino to laptop via USB
 - Observe motor rotation and direction as defined in the code
- 7. Test Movement on Rail Structure (Figure 8)
 - Confirm stepper motor drives the linear rail
 - Adjust code parameters (delay, direction logic) if needed
 - Fix distance at 30cm and at 1600 pulse per revolution and calculate time

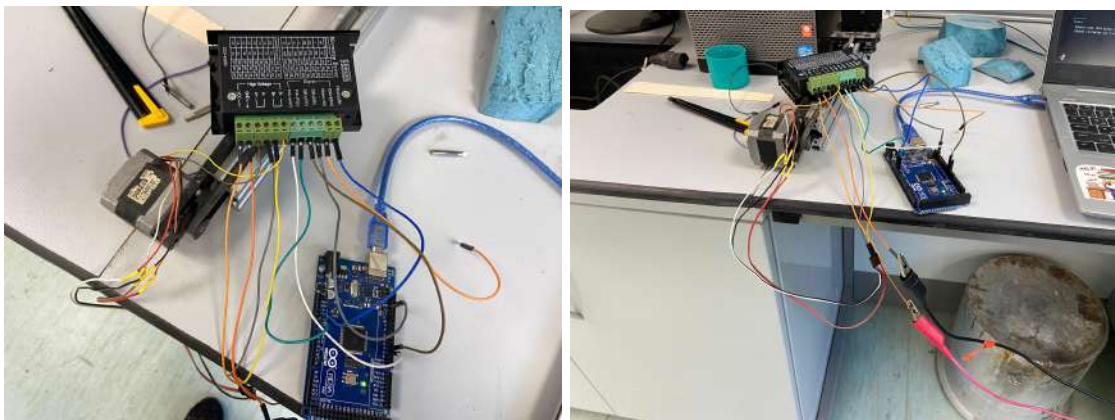


Figure 6,7: Wiring Overview

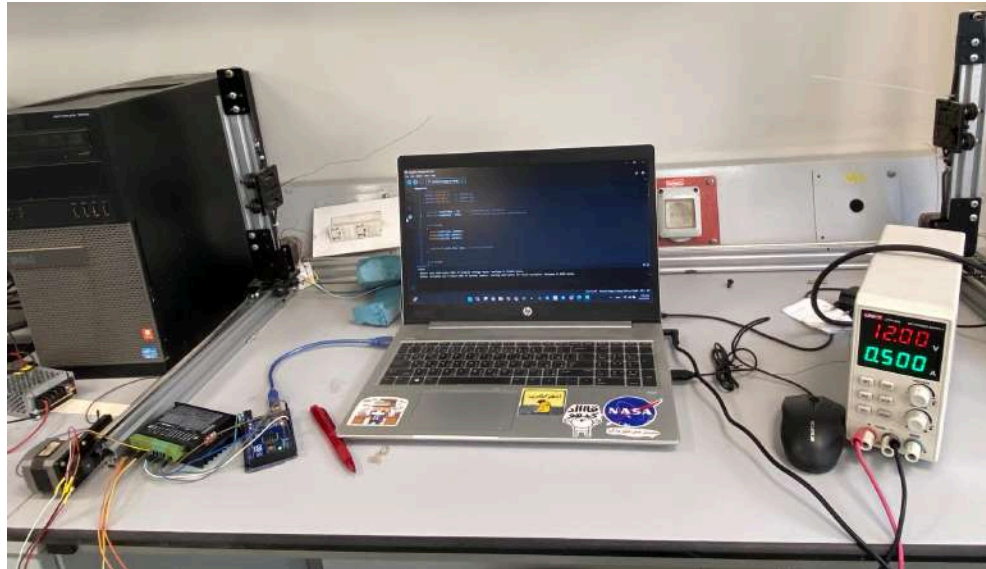


Figure 8: Full Setup

4. Results

This section evaluates the system's performance by measuring how delay times (in microseconds) between step pulses affect motor velocity and movement time.

Delay (MicroSecond) ▾	Distance (cm) ▾	#	Time ▾	#	Velocity (cm/sec) ▾
100	30	3.16			9.493670886
200	30	5.19			5.780346821
400	30	10.11			2.96735905
800	30	19.73			1.520527116
1000	30	24.75			1.212121212
2000	30	48.96			0.612745098
4000	30	97.32			0.3082614057
6000	30	146.12			0.2053107035

Table 2: Stepper Motor Performance at Varying Delays

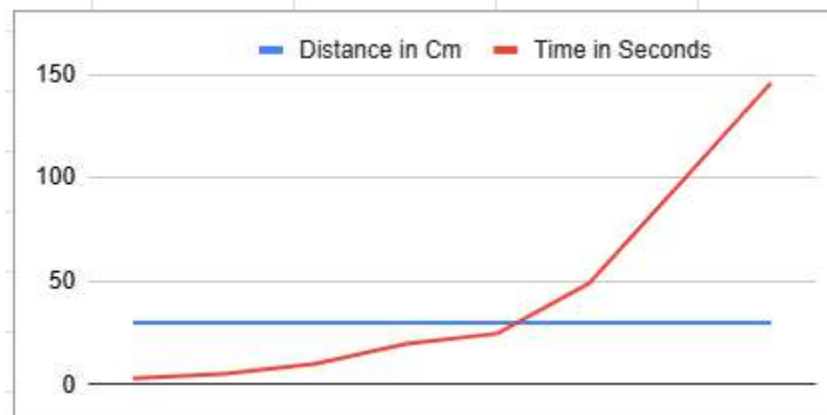


Figure 9: Distance and Time vs. Delay

Comments

- Blue curve shows constant travel distance (30 cm)
- Red curve shows increasing time with larger delays
- Delay directly increases duration for the same distance

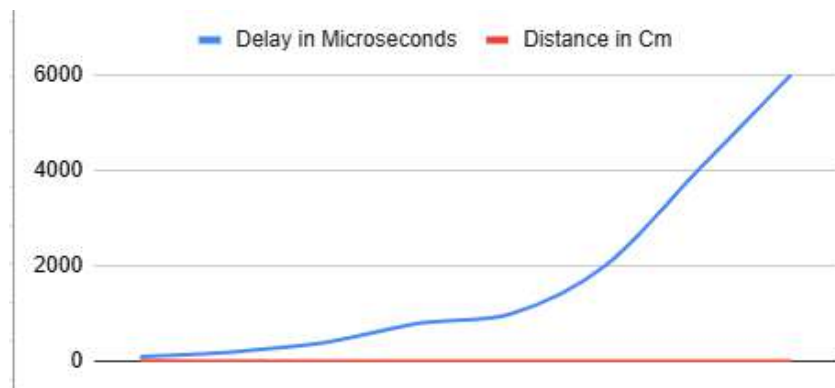


Figure 10: Delay vs. Distance

Comments

- Delay increases sharply while distance remains fixed
- Confirms that time is the only changing output for a set travel

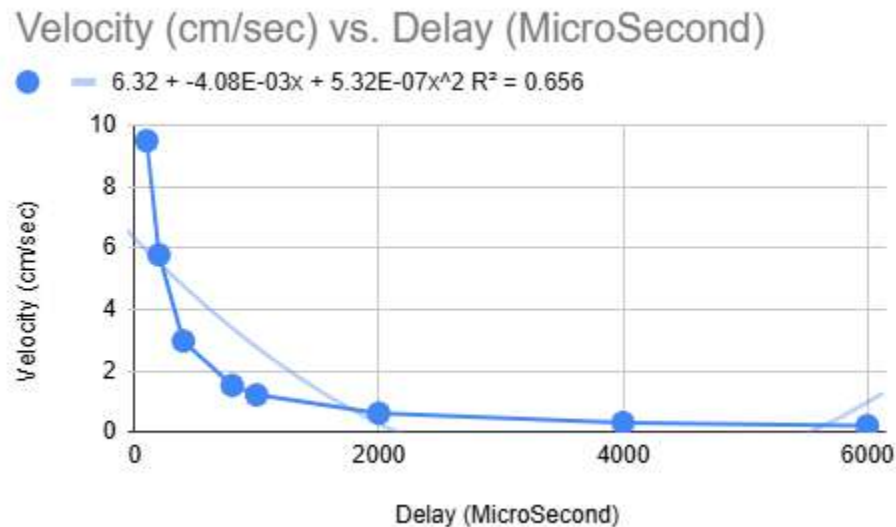


Figure 11: Velocity vs. Delay

Comments

- Velocity drops exponentially as delay increases
- At 100 μ s delay, speed reaches 9.49 cm/sec
- At 6000 μ s, speed drops to 0.20 cm/sec A second-degree polynomial trendline has been fitted to the velocity data
- The displayed equation:

$$v = 6.32 - 4.08 \times 10^{-3} \cdot x + 5.32 \times 10^{-7} \cdot x^2$$

- The R^2 value = 0.656, indicating a moderate fit
- The curve shows an initially steep drop in velocity followed by a flattening response as delay increases
- Confirms a non-linear relationship between step delay and linear velocity
- Confirms inverse relationship between delay and velocity

5. Future Work

A proposed extension of this project involves controlling four stepper motors simultaneously:

- Two motors will operate along the horizontal axis
- Two motors will operate along the vertical axis
- All motors will move in parallel, synchronized by software

This setup will differ in multiple aspects:

- Wiring: Additional connections for DIR and PUL signals
- Code: Arduino code must control four independent step signals with correct timing
- Drivers: Either one driver per motor or shared drivers depending on current rating and wiring
- Controller: Evaluation needed to determine if a single Arduino Mega can handle all control signals or if a second microcontroller is required

This design aims to expand the foam laser cutter into a full 2D motion system with coordinated axis control.

6. Concolusion

The implemented system successfully ran a stepper motor with consistent and controllable motion using an Arduino Mega and TB6600 driver. The setup proved stable and functional across various delay settings, with velocity decreasing non-linearly as delay increased. All components operated reliably, and the velocity vs. delay curve confirmed expected behavior for open-loop stepper motor control.

Key outcomes:

- Wiring and signal configuration were correctly established for the selected driver and motor
- Arduino successfully controlled motion using simple digital signals
- Velocity trends were consistent and predictable, supported by experimental data
- Polynomial fitting produced an equation to model motor behavior for future adjustments

This project establishes a working platform for stepper-based motion. It sets the stage for future upgrades such as multi-axis synchronization and full-frame foam cutting automation.

7. Refrences

[1] Bulk-Man 3D, “TB6600 Stepper Motor Driver BM3D v1.1,” *Bulk-Man 3D*, [Online].

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[2] Astrosyn, “MY14HY7-1 Astrosyn Stepper Motor NEMA 14,” *Astrosyn International Technology Ltd*, [Online]. Available:

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[3] Lin Engineering, “Stepper Motor Wiring Diagrams,” *Lin Engineering*, [Online]. Available:

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8. Appendix

8.1. Driver Datasheet

Electrical Specification:

- Input Current: 0-5.0A
- Output Current: 0.5-4.0A
- Power (MAX): 160W
- Micro Step: 1, 2/A, 2/B, 4, 8, 16, 32
- Temperature: -10~45°C
- Humidity: No Condensation
- Weight: 0.2 kg
- Dimensions: 96x56x33 mm

Figure A1: Electrical Specifications of the TB6600 Stepper Motor Driver

Two-phase 4-wire, 6-wire, 8-wire motor wiring, as shown below:

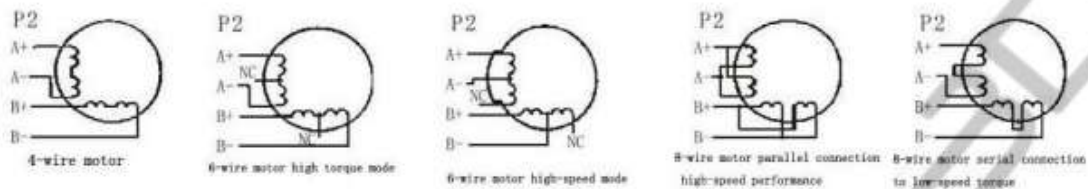


Figure A2: Wiring Diagrams for 4-Wire, 6-Wire, and 8-Wire Stepper Motors

Micro Step	Pulse/Rev	S1	S2	S3
NC	NC	ON	ON	ON
1	200	ON	ON	OFF
2/A	400	ON	OFF	ON
2/B	400	OFF	ON	ON
4	800	ON	OFF	OFF
8	1600	OFF	ON	OFF
16	3200	OFF	OFF	ON
32	6400	OFF	OFF	OFF

Table A1: Microstep Configuration Table for TB6600 Driver

Current (A)	S4	S5	S6
0.5	ON	ON	ON
1.0	ON	OFF	ON
1.5	ON	ON	OFF
2.0	ON	OFF	OFF
2.5	OFF	ON	ON
2.8	OFF	OFF	ON
3.0	OFF	ON	OFF
3.5	OFF	OFF	OFF

Table A2: Current Setting Table for TB6600 Driver

8.2. Arduino code

```
// Pin definitions
#define STEP_PIN 3 // TB6600 PUL+
#define DIR_PIN 4 // TB6600 DIR+
#define LED_PIN 13 // Onboard LED

const int totalSteps = 10; // Microsteps per revolution
const int stepDelay = 6000; // Delay between microsteps 1600 pulse/rev.
```

```
void setup()
{
    pinMode(STEP_PIN, OUTPUT);
    pinMode(DIR_PIN, OUTPUT);
    pinMode(LED_PIN, OUTPUT);

    digitalWrite(DIR_PIN, LOW); // Initial direction
}

void loop()
{
    // Turn on LED
    digitalWrite(LED_PIN, LOW);

    // Step motor forward
    for (int i = 0; i < totalSteps; i++)
    {
        digitalWrite(STEP_PIN, HIGH); digitalWrite(LED_PIN, HIGH);
        delayMicroseconds(stepDelay);
        digitalWrite(STEP_PIN, LOW); digitalWrite(LED_PIN, LOW);
        delayMicroseconds(stepDelay);
    }
}
```